Acta Oecologica 78 (2017) 47–52

Contents lists available at ScienceDirect

Acta Oecologica

journal homepage: www.elsevier.com/locate/actoec

Forum

Nest-niche differentiation in two sympatric columbid species from a Mediterranean *Tetraclinis* woodland: Considerations for forest management

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ARTICLE INFO

Article history:
Received 11 August 2016
Received in revised form
1 December 2016
Accepted 8 December 2016
Available online 3 January 2017

Keywords:
Niche
Tree occupancy
Forest management
Columba palumbus
Streptopelia turtur
Morocco

ABSTRACT

Studies of niche partitioning among Columbid species have mainly addressed food habits and foraging activities, while partitioning in relation to nest-niche differentiation has been little studied. Here we investigate whether two sympatric columbid species—Woodpigeon (*Columba palumbus*) and Turtle dove (*Streptopelia turtur*)—occupy similar niches. A total of 74 nests were monitored: 37 nests for each species. The study, conducted in June 2016, attempted to determine the factors that may play a role in nest-niche differentiation among the two sympatric columbid species in a Moroccan Thuya (*Tetraclinis articulata*) forest. We used Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) to test the relevance of nest placement, proximity of food resources, forest edge and human presence variables in the nest distribution of the two species. The results show substantial niche segregation in the *T. articulata* nest-trees selected by Woodpigeons and Turtle doves, with selection depending primarily on the tree size and nest height. Observed nest-niche partitioning may diminish the potential for competition between these species and enhance opportunities for their coexistence. Management policies and practices aimed at ensuring the presence of mixed-sized class of Thuya trees must be prioritized. We recommend additional studies designed to: (1) reproduce the same experimental approach on other Mediterranean Thuya forests to improve our understanding of the effects of different levels of anthropogenic disturbance on the breeding behaviour of these two game species; (2) better understand the spatio-temporal dynamics of Woodpigeon and Turtle dove coexistence in the region; and (3) better identify the spatio-temporal extent of the effect of forest management on Woodpigeon and Turtle dove site occupancy.

1. Introduction

Forest management that influences habitat of threatened wildlife species has been one of the principal challenges to forest managers and wildlife biologists for the last several decades (Stephens et al., 2014; Rodriguez et al., 2016). To understand the relationships between forest management and bird populations, researchers need to relate forest management to avian communities, and explore relationships at temporal and spatial scales. Identifying and understanding the mechanisms that link forest management to bird population processes will help scientists sustain species that use forest ecosystems (Marzluff et al., 2000; Gram et al., 2003).

The selection of nest location is considered the main adaptive response to nest predation pressure (Yanes and Oñate, 1996; Yanes et al., 1996; Penloup et al., 1997; Mezquida and Marone, 2002; Barrientos et al., 2009), to climatic conditions (Ferguson and Siegfried, 1989) or to human disturbances (Rodgers, 1996; Hanane and Baâmal, 2011; Hanane, 2012, 2014b).

The coexistence of similar species that share resources in the environment is also of central interest in community ecology (Begon et al., 2006; Atiénzar et al., 2013; Hanane, 2015). These species could harvest different ecological resources and/or their activity patterns could be segregated in time (Tokeshi, 1999). Species that have similar requirements may thus coexist in the same habitat if one of them adjusts its niche use either by direct behavioural mechanisms or at a longer time-scale by the displacement of one or more morphological, ecological, or physiological characters through evolutionary adaptation (Brown and Wilson, 1956; Moreno and Carrascal, 1993; Martin and Thibault,

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http://dx.doi.org/10.1016/j.actao.2016.12.003

1996; Atiénzar et al., 2013; Lapiedra et al., 2013). In columbid species, studies of niche partitioning have mainly addressed food habits and foraging activities (Wolf et al., 2002; Sol et al., 2005; Hayslette, 2006) in addition to vocal activities (de Kort et al., 2002; Kopij, 2003). Little is known about the relative effects of habitat characteristics and human factors on nest-site selection of columbid species (however, see Diamond, 1973, 1975; Ruiz, 2012 and Hanane, 2015) both in Morocco and in Euro-Mediterranean countries.

The present study analyses the patterns of nest habitat selection of two sympatric granivorous birds to examine whether they are segregated on some nest site characteristics despite their apparent similarity in terms of breeding ecology. More specifically we studied nest location in two columbid species (Aves, Columbidae, Columbinae): the Woodpigeon (*Columba palumbus*; Linnaeus, 1758) and the Turtle dove (*Streptopelia turtur*; Linnaeus, 1758) in an agroforestry land use system where they breed in Thuya forest *Tetraclinis articulate* (Vahl) Mast. The two species are part of a well-defined ecological guild of columbid birds linked to lowland cereals during spring and summer (Buckingham et al., 2010). The choice of these species is dictated by: (i) their cynegetic importance; and (ii) their significant presence is the studied forest area. They are major game species in Morocco, highly valued by national and international hunters (HCEFLCD, 2013). The coexistence of these two breeding columbids has also been reported in the Middle Atlas Mountain, in natural Holm oak trees (*Quercus rotundifolia*) (Hanane, 2014a).

The Turtle dove is a sub-Saharan migratory bird that has an extensive breeding distribution range in the western Palearctic (Cramp, 1985). In its European breeding areas, the species underwent a rapid and serious decline across its European range (−78% 1980–2013; PECBMS, 2015; Dunn et al., 2016) and has been classified as 'Vulnerable' throughout Europe ('Near Threatened' within the EU27 countries) following a recent assessment (BirdLife International, 2015). The major threats to the Turtle dove appear to be nesting habitat degradation (Browne et al., 2004), changes in food availability (Browne and Aebischer, 2003) and agricultural land use in addition to hunting (Boutin and Lutz, 2007). Other factors that may contribute to the decline of Turtle doves are changes in wintering grounds and changes associated with migration (Browne and Aebischer, 2001; Bakaloudis et al., 2009; Eraud et al., 2009). In North Africa, the species is known to be a common migratory breeding bird (Isenmann and Moali, 2000; Thévenot et al., 2003; Isenmann et al., 2005). In Morocco, the Turtle dove is mostly found in agricultural landscapes especially in irrigated areas where fruit orchards are broadly represented (Hanane and Besnard, 2014).

The Woodpigeon breeds across the European continent, from Russia and Scandinavia to the Atlantic coasts and Mediterranean peninsulas (Saari, 1997), and in North Africa (Isenmann and Moali, 2000; Isenmann et al., 2005). Despite being a game species, woodpigeons belong to the NON SPEC^{E} category in Europe (Concentrated in Europe but with a Favourable Conservation Status) (Tucker and Heath, 1994). In Morocco, this subspecies is mainly widespread and abundant in mountainous areas (Middle, High and Anti-Atlas) (Thévenot et al., 2003). Woodpigeons are known to be common residents and winter visitors (Thévenot et al., 2003). In Morocco, the breeding population is mainly sedentary. The woodpigeons actively defend the territory borders during the breeding season while occupying a variety of habitats, including forests of oaks, cedar, pines and conifers (Thévenot et al., 2003).

The use of the same habitats by two-species can reduce or extirpate other populations by niche displacement through competition, alteration of nutrient cycles, hybridization, and introgression. Thus, the major aims of this study were to (1) identify the most important factors affecting the distribution of nests of the two columbid species in a Moroccan agroforestry area where they co-occur; (2) assess if there is an overlap in microhabitat use for nesting and, in the case of no overlap, (3) evaluate the nature of nest-niche differentiation among these two coexisting columbid species.

2. Materials and methods

2.1. Study area

Observations were made in a Moroccan natural Thuya forest (32.8 km^2) located within Oued Mellah watershed at the vicinity of Ben ahmed city (Fig. 1). Under semi-arid climate, it receives 320 mm of mean annual rain, most of which falls during the winter rainy season (November–January). Temperature varies widely, being more temperate during the winter, but with peaks in summer reaching, at times, 42 °C. The altitude is ranging from 450 to 649 m. In addition to the Thuya forest, the landscape is also composed of cereal crops (600 ha), vegetated ravines (250 ha) and olive plantations (200 ha). The tree-layer consists mainly of Thuya in association with wild olive (*Olea europaea oleaster*) and Tizra tree (*Rhus pentaphylla*), while cereal crops are dominated by wheat (*Triticum turgidum and T. aestivum*) and barley (*Hordeum vulgare*). Vegetated ravines are dominated by the common rush (*Juncus effuses*), Oleander (*Nerium oleander*), and Chilean myrtle (*Luma apiculata*). In this landscape, olive orchards are distributed in very small patches (1.52 ± 0.05 ha) throughout the area. Very small and single-family houses, most distant from each other (2.10 ± 0.85 km), are found in the vicinity of crops and forests. In the study area, there is no human habitation inside the forest area (HCEFLCD, 2006).

2.2. Sampling design

Given that the Thuya forest is the sole habitat within which the two columbid species breed in sympatry, fieldwork was exclusively carried out in this forest habitat type. The monitoring took place in June, during the breeding season, in 2016. This period was chosen because most pairs of both species are nesting at this time (Thévenot et al., 2003). The forest study zone was initially gridded with square 100 × 100 m cells (20.000 m^2 areas). The sampling unit

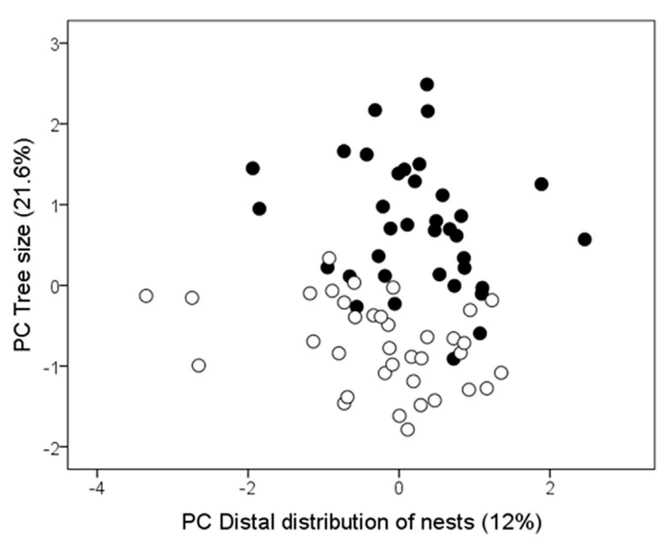

Fig. 1. Distribution of the 74 sampled nests in $PC_{Tree\ size}$-$PC_{Distal\ distribution\ of\ nests}$ space derived from the PCA of the transformed original variables. Black circles represent Woodpigeon nests and open circles represent Turtle dove nests.

size was considered to increase the possibility to find nests of both species. Each grid cell was assigned a unique numerical identifier. 15 cells were randomly chosen by drawing 15 numbers using a random number generator. Within each cell, we systematically searched to locate nests. Because the peak of breeding (Thévenot et al., 2003) differed significantly between the two species in Morocco, we have chosen to have a balanced design (i.e. 37 nests for each species).

The surveys began 1 h after sunrise (min. Value) and lasted until 1 h before sunset (max. Value) under favourable weather conditions (no rain and no or low wind). We systematically searched to locate nests in all sampled grid cells. A nest was considered active when eggs, nestlings, or incubating adults were present. The location of each nest tree was referenced using a portable GPS (Magellan eXplorist XL), and then reported using Quantum GIS v1.7.3 (QGIS).

2.3. Environmental variables

To describe nest-niches, we focused on twelve variables: two variables for nest position [nest height (in m) above ground (NH) and nest-trunk distance (NT)], two variables for tree size [nest tree (in m) height (TH) and diameter (in m) at breast height (DBH)], three variables for human activities [closest distance from each nest-tree (in m) to track (DTR), and to habitation (DNH)], two variables for vegetation structure [vegetation cover (VC) and density of trees (DT) within 5 m radius circle centered on nest-tree], two variables for proximity to food resources [closest distance from each nest-tree (in m) to cereal crops (DNC) and to water point (DW)], one variable for edge effect [closest distance from each nest-tree (in m) to forest edge (DFE)] and one variable for topography [altitude (ALT)].

When there were no species in a nest when located, the two variables for nest position (NH and NT) and tree size (TH and DBH) were immediately measured using a clinometer and diameter tape. If one of the species was present, these measurements were taken early in the morning, when birds often leave the nest to look for food. Variables related to human disturbance (DNC and DNH), proximity to food resources (DNC and DW) and forest edge were measured with QGIS.

2.4. Statistical analysis

Before performing statistical analyses, we checked for normality and homogeneity of variance for all the variables. Variables that did not conform to the requirements for parametric tests were log-transformed prior to all analyses (Zar, 1984; Underwood, 1996; Quinn and Keough, 2002). We also checked for possible correlations among variables by using Pearson's rank correlation (r) index. We used the Principle Component Analysis (PCA) because it is a powerful method for analysing ecological niches (Janzekovic and Novak, 2012). We thus collapsed habitat structure variables into independent vectors using (PCA), since this analysis allowed us to: (1) reduce the dimensionality of the set of variables to a smaller number of ‘representative’ and ‘uncorrelated’ variables ($n = 4$); (2) investigate multicollinearity; and (3) describe dominant ecological gradients (Legendre and Legendre, 1998). For each PCA, a varimax normalized rotation was applied to the set of principal components with eigenvalues >1.0 to obtain simpler and more interpretable gradients (Legendre and Legendre, 1998). We interpreted the biological meaning of the principal components, which explain the greatest amount of combined variation within the habitat structure data, by examining the component loadings of each variable (McGarigal et al., 2000). After this first stage of analysis, we performed Linear Discriminant Analysis (LDA) on the PCA axes to calculate the discriminant function. The overall correct classification rate was also evaluated.

All statistical analyses were performed with SPSS software (version 15.5). A two sample *t*-test was used to determine whether nest placement varied between *Columba palumbus* and *Streptopelia turur*. Means are quoted ± standard errors. Results were considered significant at $P < 0.05$. To account for the two sample multiple *t*-tests, we used the Bonferroni correction (Rice, 1989), and considered differences to be significant when $P < 0.0041$.

3. Results

The *t*-tests conducted on nest placement characteristics of *C. palumbus* and *S. turtur* demonstrated that tree height (TH), nest height (NH) and diameter at breast height (DBH) were significantly different, while altitude (ALT), nest-trunk distance (NT), vegetation cover (VC), tree density (TD), closest distance from each nest-tree (m) to track (DTR), to habitation (DNH), to cereal crops (DNC), to water point (DW) and to forest edge (DFE) were not (Table 1).

To test the relevance of nest placement, human disturbance and proximity to food resources variables in Woodpigeon and Turtle dove nest distributions, we performed a PCA. The PCA summarized the studied field variables into four independent axes accounting for 72.7% of the total variance of the original dataset. PC1 and PC2 were considered as the most.

important axes, explaining 24.5% and 21.6% of the variance, respectively (46.1% of the total variance across all measured variables).

The varimax rotation revealed a first gradient ($PC_{\text{Human disturbance}}$) characterized by high loadings of variables related to human presence, given its positive correlation with distance from nest to habitation ($r = 0.690$, $P < 0.001$), to cereal crops ($r = 0.902$, $P < 0.001$), to tracks ($r = 0.683$, $P < 0.001$) and to forest edge ($r = 0.859$, $P < 0.001$). The second gradient ($PC_{\text{Tree size}}$) represents an axis of increasing height of trees with high DBH, as it was positively correlated with tree height ($r = 0.910$, $P < 0.001$) and DBH ($r = 0.884$, $P < 0.001$). The third gradient ($PC_{\text{Proximity to water}}$) described gradients of increasing distance to water points ($r = 0.614$, $P < 0.001$) and the fourth axis ($PC_{\text{Distal distribution of nests}}$) expressed decreasing distal distance of nests, given that it was negatively correlated with nest-trunk distance ($r = -0.828$, $P < 0.001$). The Kaiser-Meyer-Olkin of sampling adequacy (KMO) indicated that our data were suitable for the PCA (KMO = 0.685; Bartlett-test for sphericity, $\chi^2 = 330.97$, $P < 0.001$).

Detailed examination of the individual nests on $PC_{\text{Tree size}}$-$PC_{\text{Distal distribution of nests}}$ space shows that both colombidae species are clearly separated on the first axis, with *C. palumbus* using respectively higher values of TH and DBH and *S. turtur* using lower values of TH and DBH (Fig. 1). The second axis also separates *C. palumbus* from *S. turtur*; woodpigeons had lower values for nest-trunk distance than did Turtle doves (Fig. 1).

We used an LDA to test which of the four PCA axes maximizes variation in nest-niche between the two colombidae species. This analysis showed the relevance of the $PC_{\text{Tree size}}$ axis and the $PC_{\text{Distal distribution of nests}}$ in discriminating between the two columbid species (Tables 2 and 3). The prediction of the classification using the linear discriminant function is very good since 90.3% of the original observations were correctly classified.

4. Discussion

The results of the analyses support the existence of a substantial niche separation in Thuya nest-trees selected by the Woodpigeon and the Turtle dove in relation to the tree size.

The Woodpigeon chose nesting trees that were greater in height

Table 1
Untransformed values of the environmental variables measured at Woodpigeon (n = 37) and Turtle dove nests (n = 37) in a *Tetraclinis* woodland, Morocco, 2016. An asterisk (*) indicates that data differed significantly (two-sample t-tests, $P < 0.05$).

Variables	Columbidae species		*t*-value	*P*-value
	Columba palumbus	*Streptopelia turtur*		
Altitude (m)	455.14 ± 8.64	474.22 ± 8.89	−1.540	0.128
Nest tree height (m)	8.45 ± 0.26	6.17 ± 0.13	8.123	<0.004*
Nest height above ground (m)	4.95 ± 0.20	2.86 ± 0.11	7.346	<0.004*
Diameter at breast height (cm)	58.69 ± 2.07	49.44 ± 1.13	2.902	<0.004*
Vegetation cover (%)	10.94 ± 2.08	18.06 ± 3.42	−1.778	0.081
Tree density	3.28 ± 0.16	2.89 ± 0.18	1.596	0.115
Distance from nest to trunk (cm)	27.86 ± 5.18	46.81 ± 7.47	0.778	0.439
Distance from the nest-tree to the nearest habitations (m)	869.68 ± 35.96	806.62 ± 34.51	1.265	0.210
Distance from the nest-tree to the nearest track (m)	316.03 ± 28.52	299.50 ± 30.83	0.393	0.695
Distance from the nest-tree to the nearest cereal crops (m)	454.75 ± 33.01	407.94 ± 32.75	1.006	0.318
Distance from the nest-tree to the nearest water points (m)	856.17 ± 35.89	1231.47 ± 133.42	−1.461	0.151
Distance from the nest-tree to forest-field edge (m)	375.47 ± 29.72	296.69 ± 31.06	1.832	0.071

The 0.05 alpha level is adjusted to 0.004 using a Bonferroni correction.

and larger in DBH compared to Turtle doves. This finding is in agreement with Hanane et al. (2012) who have reported that breeding Woodpigeons often select trees that are distinguished by their height and their DBH in the Holm oak (*Quercus rotundifolia*) Middle Atlas Mountain forests. The attraction of Woodpigeons for taller trees has often been reported (Aubineau and Boutin, 1998; Rouxel and Czajkowski, 2004), although this has been virtually unstudied quantitatively. Tall trees may permit escape from human disturbance (Dutta, 2007; Hanane and Baamal, 2011) and may also provide a distinct landmark to help commuting adults in locating the nest (Silvergieter and Lank, 2011). Our findings are consistent with the existing literature testifying to the role of the Woodpigeon as an indicator species of the forest stand height (Regnery et al., 2013; Hanane et al., 2012; Hanane, 2014b).

Statistical analyses showed that Turtle doves avoid large Thuya's trees. This is not surprising since the Turtle dove is known for using a medium-sized class of forest trees. One possible explanation of the nest-niche segregation between these species is competitive interactions between Turtle doves and Woodpigeons. In competitive interactions, larger species (the case of the Woodpigeon) are typically dominant over smaller ones (the Turtle dove), excluding the smaller species from the landscape or at least reducing their density (Rosenzweig, 1966; Wereszczuk and Zalewski, 2015). In addition, and alternatively, with a smaller weight and size, Turtle doves seem not to be able to tolerate a too enclosed microhabitat (Dunn and Morris, 2012; Hinsley et al., 1995), at least compared to Woodpigeons. Medium-sized class of conifer trees seems thus suitable for Turtle doves (Bakaloudis et al., 2009), by providing a less enclosed canopy allowing a good aeration, a shaded canopy for better concealment and by facilitating trips to and from nests made by both adults and fledged young doves (Hanane, 2012). Conduct further studies in areas where each of the two studied species breeds alone prove to be of a great importance. This would allow assessing the change in nest-site selection due to the presence of another breeding columbid species. Additionally, this could also confirm or disprove the presence of competitive exclusion.

Nest height also allows niche segregation between the two species. Results show that Woodpigeons occupy higher parts of the tree compared with Turtle doves, which occupy medium parts. These two different strategies have been previously documented respectively for Turtle doves by Hanane (2012, 2014a, 2015) in Moroccan agroecosystems and for Woodpigeons (Hanane et al., 2012; Hanane, 2014b) in Middle Atlas Cork oak forests. Such behaviour could be a strategy to avoid nest predation (Mitrus and Socko, 2008; Hanane, 2012), human disturbances (Hanane and Baamal, 2011; McCarthy and Destefano, 2011), to provide protection from inclement weather (Sadoti, 2008).

The proximity of food resources (water and cereals) and habitat disturbance play no role in discriminating between the two columbid species. The landscape configuration (Thuya forests bordered by tracks, cereal crops and habitations) and the use of the same nesting forest areas would explain this outcome. However, the Turtle doves longer use the forest edges than do the Woodpigeons (see Table 1, Reif et al., 2008; Terraube et al., 2016), but this spatial difference remains insignificant to draw conclusions. This species, although using the forest edges, is known to be a reliable indicator of: (i) the proximity to farmlands; and (ii) ecological state of agricultural landscapes (BirdLife International, 2015; Tucker and Evans, 1997).

In the study area, Woodpigeons and Turtle doves are primarily focused on finding suitable Thuya's nest-trees. The lack of variability among the different metrics (Table 1) suggests that the study area is fairly homogenous; suggesting that landscape-level metrics, here considered, was not an important driver in discriminating between the two species.

Table 2
Results of the discriminant function analysis between Woodpigeon (n = 37) and Turtle dove (n = 37) nest placement parameters in a *Tetraclinis* woodland, Morocco, 2016.

	Function 1
Eigenvalue	1.686
Percentage of eigenvalue associated with the function (%)	100.00
Canonical correlation	0.792
Wilks' Lambda	0.372
Chi−square statistic	67.199
Significance (degrees of freedom)	$P < 0.001$ (4)
Standardized canonical discriminant function coefficients	
$PC_{Human\ disturbance}$	0.086
$PC_{Tree\ size}$	1.028
$PC_{Water\ proximity}$	-0.160
$PC_{Distal\ distribution\ of\ nests}$	0.509

Table 3
Significant predictor variables obtained from discriminant analyses between Woodpigeon (n = 37) and Turtle dove (n = 37) nest placement parameters in a *Tetraclinis* woodland, Morocco, 2016.

	Wilks.lambda	*F*-value	*P*-value
$PC_{Human\ disturbance}$	0.998	0.122	0.728
$PC_{Tree\ size}$	0.445	87.443	< 0.001
$PC_{Proximity\ to\ water}$	0.994	0.423	0.518
$PC_{Distal\ distribution\ of\ nests}$	0.935	4.186	0.049

Generally, this work shows that for nesting, Turtle doves avoid large Thuya trees, while Woodpigeons can build nests in large ones. Thus, although they are nesting in the same forest stand, Woodpigeons and Turtle doves differ largely in their choosing of nesting trees. From a conservation point of view, this argument would allow the persistence of the two species in this agroforestry landscape and reduce competition between these two dove species.

4.1. *Considerations for forest management*

The results of the present study suggested that nest-niche segregation between Turtle doves and Woodpigeons was due to characteristics of nest-trees. In the studied Thuya forest, Woodpigeons select large trees, whereas Turtle doves choose relatively medium-small trees. We encourage and recommend forest managers to place a greater emphasis on ensuring the presence of mixed-sized class of Thuya trees. This necessarily goes through good forest practices by avoiding loss of large trees and by ensuring the presence of relatively small trees. Indeed, maintaining forests at different successional stages ensures that a diversity of niches are available and positively impacts several forest-dwelling species (Moning and Müller, 2008; Lafond et al., 2015). From a practical point of view, introducing a rotating system of forest utilization for timber harvesting which would promote forest regeneration after several years of deferred harvesting, and thus provide, at medium- and long-term, a greater availability of nesting sites for Woodpigeons and Turtle doves, while ensuring the development of both shrubs and herbaceous layers, would be extremely beneficial.

Funding

This study was supported by the Forest Research Center, High Commission for Water, Forests and Desertification Control, Morocco.

Author contributions

Project design: S. Hanane, M. Yassin.
Data collection: S. Hanane, M. Yassin
Data analyses: S. Hanane.
Writing the article: S. Hanane.
The two authors have seen and approved the final article.

Acknowledgments

We thank Abderrahim and Mustapha for helping during the field work. We also thank two anonymous reviewers and the Editor of Acta Oecologica for their comments and advice.

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